

The V2G Payback Calculator

Methodology, assumptions, and formulas

A companion note explaining how the interactive V2G Payback Calculator works. It documents the tool's inputs, outputs, formulas, and data assumptions. All figures are illustrative estimates for discussion and do not constitute financial advice. No vehicle, charger, utility, or platform is endorsed.

1. What the tool does

The V2G Payback Calculator is an interactive tool that estimates whether the revenue an electric vehicle could earn by selling power back to the grid would offset the cost of the bidirectional (two-way) charger required to participate. It is designed around a single question:

“If an EV owner allows their vehicle to feed power back to the grid, would the earnings eventually pay off the specialised charger needed to do so?”

The tool translates a general argument, that the household business case for vehicle-to-grid (V2G) is currently thin, into a figure the reader generates from their own inputs. Rather than presenting a fixed conclusion, it lets the user explore the economics for their own region and circumstances.

2. Inputs the user controls

The tool accepts four inputs, each corresponding to a real decision an EV owner would face:

Region. The user selects one of ten regions. This is significant because V2G earnings depend heavily on local market rules and programme maturity. Regions with established programmes and tariffs support higher earnings, while regions where the market is still forming support lower earnings. Region acts as a multiplier that scales the earnings estimate up or down.

EV battery size. A larger battery can offer more energy to the grid, so earnings rise as battery size increases. The relationship shows diminishing returns, reflecting that programmes do not draw on the entire battery.

Days per week of participation. How frequently the owner plugs in and permits discharge. More frequent participation increases earnings, again with diminishing returns, reflecting that programmes do not call on any single vehicle continuously.

Bidirectional charger cost. The installed cost of the two-way charger. This is the amount the estimated earnings must recover.

3. Outputs the tool displays

Once the inputs are set, the tool immediately calculates and displays the following:

Estimated annual earnings. An approximate figure for what the vehicle could earn per year. This is anchored to published 2025 pilot data, in which participating households earned a median of roughly USD 420 to 780 per year from grid services and bill savings. The tool begins from this range and adjusts it by region, battery size, and participation.

Effective battery utilisation. The share of the year during which the vehicle actually feeds the grid. Reported figures indicate that even active participants discharge to the grid less than 2 percent of the year. The tool presents this as a percentage and a bar to illustrate that the battery remains idle for the vast majority of the time.

Discharge hours per year. The same utilisation expressed in hours, typically fewer than 200 per year.

Payback period. The headline figure, calculated by dividing the charger cost by the estimated annual earnings. For example, a charger costing USD 4,500 that earns USD 630 per year implies a payback of about 7 years.

Written verdict. A short interpretation that changes with the result: under 5 years is described as workable but tight; roughly 5 to 10 years as marginal; and beyond that as a case that does not close, noting that established real-world models have tended to rely on public subsidy or fleet economics rather than individual households.

4. The formulas

The calculations use the following variables:

- R = regional multiplier (set by the region selected)
- B = battery size in kWh
- D = days per week of participation (1 to 7)
- C = bidirectional charger cost in USD
- BASE = 600 (the midpoint of the USD 420 to 780 pilot range)

4.1 Battery factor

Scales earnings by battery size, with diminishing returns. A 60 kWh battery yields a factor of 1.0 (the baseline).

$$\text{batteryFactor} = (B / 60) ^ 0.5$$

4.2 Participation factor

Scales earnings by participation frequency, also with diminishing returns. Four days per week yields a factor of 1.0 (the baseline).

$$\text{daysFactor} = (D / 4) ^ 0.6$$

4.3 Annual earnings

The baseline is adjusted by region, then by battery size and participation.

$$\text{annualEarnings} = \text{BASE} \times R \times \text{batteryFactor} \times \text{daysFactor}$$

4.4 Utilisation

Utilisation rises with participation but is capped below 2 percent, consistent with reported V2G utilisation figures.

$$\text{utilisationPercent} = \text{minimum of } (1.9 , 0.5 + D \times 0.20)$$

4.5 Discharge hours per year

Utilisation converted to hours, based on 8,760 hours in a year.

$$\text{dischargeHours} = (\text{utilisationPercent} / 100) \times 8760$$

4.6 Payback period

The charger cost divided by the estimated annual earnings.

$$\text{paybackYears} = C / \text{annualEarnings}$$

5. Regional multipliers

The regional multiplier reflects how developed each market's V2G rules and tariffs are, relative to a baseline of 1.00. Higher values indicate more established monetisation pathways.

Region	Multiplier (R)
EU / EMEA	1.25
ANZ	1.15
North America	1.05
Greater China	1.00
JPAC / Japan	0.95
APAC (broad)	0.80
Middle East	0.70
Southeast Asia	0.65
Latin America	0.60
Africa	0.50

6. Worked example

Using North America, a 60 kWh battery, 4 days per week of participation, and a USD 4,500 charger:

Input	Value
Region multiplier (R)	1.05 (North America)
Battery size (B)	60 kWh
Participation (D)	4 days per week
Charger cost (C)	USD 4,500

Step by step:

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batteryFactor = (60 / 60) ^ 0.5 = 1.0 daysFactor = (4 / 4) ^ 0.6 = 1.0 annual = 600
× 1.05 × 1.0 × 1.0 = 630 per year utilisation = min(1.9 , 0.5 + 4 × 0.20) = 1.3% hours
= 1.3 / 100 × 8760 ≈ 114 hours payback = 4500 / 630 ≈ 7.1 years

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These are the values the tool displays for this combination of inputs.

7. How the results tend to read

Across most input combinations, the estimated payback period is long, frequently seven years or more, and in some cases considerably longer. This follows directly from the arithmetic: a charger costing several thousand dollars, set against earnings of a few hundred dollars per year, takes an extended period to break even.

The tool is structured so that the user reaches this observation through their own inputs rather than being told it. The pattern it illustrates is that the underlying technology functions, while the private business case for the hardware, on its own, often does not close under current conditions.

8. Scope, data, and limitations

The tool is a deliberately simple and transparent approximation, not a peer-reviewed engineering model. Its exponents and multipliers are calibrated to reflect widely reported patterns, diminishing returns, regional variation, sub-2-percent utilisation, and long payback periods, rather than to predict any individual outcome.

- Earnings are modelled from published 2025 pilot ranges (median of approximately USD 420 to 780 per year).
- A regional multiplier scales that baseline according to the maturity of each market's V2G rules and tariffs.
- Earnings scale with battery size and participation days, with diminishing returns.
- Utilisation is anchored below 2 percent, consistent with reported findings that active participants discharge fewer than about 200 hours per year.
- Actual results depend on the user's utility, tariff, vehicle, charger, and local interconnection rules.

All figures are illustrative estimates presented for discussion and do not constitute financial advice. Ranges are drawn from the International Energy Agency and published pilot data. This document and the tool it describes form part of Grid Intelligence, an independent, vendor-neutral publication on utility OT and grid modernisation. No employer, vehicle, charger, utility, or platform is endorsed.